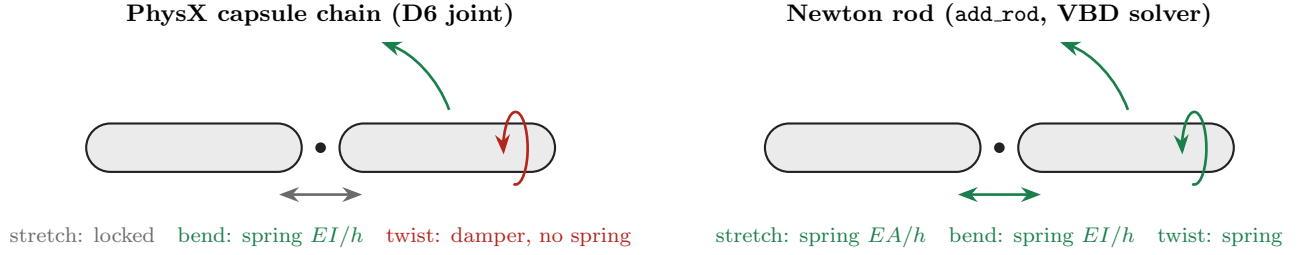


Cable models compared: Newton rod and PhysX capsule chain

Thin cable ($d = 4\text{ mm}$), and a thick cable ($d = 30\text{ mm}$) where volumetric FEM also works

The difference in one line. Both models are a chain of capsules. They differ in what the joint between two capsules can do: the capsule chain has springs for *bending only*; the Newton rod has springs for *stretch, bend and twist*, all derived from the material.



1. What each joint carries

	Capsule chain (our script)	Newton rod 1.2.0 (our script)	Newton rod, current docs
Stretch (force along the cable)	locked: links are rigid, no axial elasticity	spring EA/h	spring (stretch slot)
Bend (moment)	spring EI/h on two axes	spring EI/h	spring (bend slot)
Twist (torque about the axis)	stiffness 0, damping only	carried by the joint's angular spring	its own spring (twist slot)
Shear	locked	not modelled	spring (shear slot)
Bending hysteresis	none	none	optional Dahl friction
Time integration	PhysX iterative, 1/480 s step	VBD, implicit	VBD, implicit

2. Calculations, thin cable $L = 893\text{ mm}$, $r = 2\text{ mm}$, $E = 80\text{ MPa}$, $\nu = 0.45$, 150 segments.

segment length	$h = L/150 = 893/150 = 5.95\text{ mm}$	stretch spring	$EA/h = 1.69 \times 10^5\text{ N/m}$
area	$A = \pi r^2 = 1.26 \times 10^{-5}\text{ m}^2$	bend spring	$EI/h = 0.169\text{ N m/rad}$
second moment	$I = \pi r^4/4 = 1.26 \times 10^{-11}\text{ m}^4$	twist spring	$GJ/h = 0.116\text{ N m/rad}$
stretch rigidity	$EA = 1005\text{ N}$	capsule chain twist	0 (should be 0.116)
bend rigidity	$EI = 1.01 \times 10^{-3}\text{ N m}^2$	capsule chain stretch	rigid (should be 1.69×10^5)
twist rigidity	$GJ = EI/(1 + \nu) = 6.9 \times 10^{-4}\text{ N m}^2$		

3. Measured in our benchmark three supports, 150 segments, run `apple_cable_3pt_150`, `metrics.csv`.

	error vs catenary	error vs photo	length drift	settled	wall time
Newton rod (point pin)	0.9 mm	75 mm	0.86 %	yes, 1.1 s	12.9 s
Newton rod (tape support)	21 mm	65 mm	1.06 %	yes, 1.0 s	11.7 s
PhysX capsule chain	74 mm	33 mm	0.46 %	no, after 8.1 s	29.0 s

The test is a planar hang: it exercises weight and bending, not twist. The twist comparison above comes from the joint definitions, not from a measurement.

4. Pros and cons, thin cable

Newton rod

- + Torque and twist are transmitted elastically along the cable.
- + Tension comes from EA , so axial force is physical.
- + Stiffness set directly from E , r and segment length.
- + Implicit solver: stable at high stiffness, settles in about 1 s.
- + Matches the mathematics to 0.9 mm.
- + GPU-batched and differentiable (Warp); couples two-way with rigid-body solvers.
- Through Isaac Sim 6 USD it cannot hold both ends (articulation must be a tree), so it runs standalone.
- Holding a body also holds its angle: no true pin.
- API still changing between versions.
- Largest length drift here (0.86 %).

PhysX capsule chain

- + Native Isaac Sim: USD, Isaac Lab, ROS 2, sensors, contact with robots.
- + Both ends fixed works directly.
- + Closest to the photographed cable (33 mm).
- + Smallest length drift (0.46 %), because links are rigid.
- No twist spring in our setup: a torque spins the cable instead of loading it.
- No axial elasticity: tension is a constraint force, not $EA\epsilon$.
- 150 rigid links solved iteratively: not settled after 8 s, slowest run.
- 74 mm from the catenary.
- Not differentiable.

5. Thick cable, $d = 30$ mm: all three models work same length and material.

		Capsule	Newton rod	FEM
aspect ratio	$L/d = 893/30 = 30$			
stretch rigidity	$EA = 56.5 \text{ kN}$	Bending	yes	yes
bend rigidity	$EI = 3.18 \text{ N m}^2$	Stretch	no	yes
twist rigidity	$GJ = 2.19 \text{ N m}^2$	Twist / torque	no spring	yes
weight	$w = \rho Ag = 16.6 \text{ N/m}$	Cross-section	no	yes
bending length	$(EI/w)^{1/3} = 0.58 \text{ m}$	squeeze, contact pressure		
segments, $h = d$	$893/30 \approx 30$	True geometry, no rescaling	yes	yes
FEM voxels across	$30/6.9 = 4.4$ at $N = 130$	Unknowns to solve	30 links	≈ 2000 voxels
FEM voxel count	$\frac{\pi}{4} 4.4^2 \cdot 130 \approx 2000$		30 links	

At 30 mm the cable is no longer a thin string: bending length 0.58 m is comparable to its 0.89 m length, so it behaves as a bent beam. Rod and capsule models stay cheap but need segments about one diameter long, so they resolve curvature with about 30 segments instead of 150. FEM costs more and is the only one that shows the cross-section deforming, for example where a gripper squeezes it.

Summary. Thin cable: use the Newton rod when twist, torque or tension matter, and the capsule chain when the cable must live inside an Isaac Sim scene; FEM is not an option. Thick cable: all three work; add FEM when the cross-section or contact pressure matters, otherwise the rod is the cheaper choice.

Sources. Joint definitions and stiffness formulas: `hang_physx_capsule.py`, `hang_newton_cable.py`, `cable_config.py` in `methods/`. Rod stretch, shear, bend and twist slots and Dahl friction: Newton documentation, `SolverVBD`. Two-way coupling: NVIDIA developer blog on Newton, 16 March 2026. FEM numbers: `docs/fem_thin_objects`.